

# EC-201: Engineering Project Management

## Project Feasibility

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## 7.1 What is a Feasibility Study? (1/2)

- A feasibility study is essentially a process for determining the viability of a proposed idea and providing a framework and direction for its development and delivery. It is a process for making sound decisions and setting direction. It is also a process which:
  - Is driven by research and analysis
  - Usually involves some form of consultation with stakeholders, community, users, etc.
  - Focuses on analyzing, clarifying and resolving key issues and areas of concern or uncertainty
  - Very often involves basic modelling and testing of alternative concepts and approaches
- There is no universal format for a feasibility study. Feasibility studies can be adapted and shaped to meet the specific needs of any given situation.

## 7.1 What is a Feasibility Study? (2/2)

- A business plan is a significant investment of time and money, you want to make sure that there are no major roadblocks facing your business idea before you make that investment. *Identifying such roadblocks is the purpose of a feasibility study.*
- A feasibility study looks at three major areas:
  - a) Market issues
  - b) Organizational/technical issues
  - c) Financial issues

## 7.2 Why Do Feasibility Studies? (1/2)

- General requirements and potential benefits of conducting feasibility study include:
  - Developing any new business venture is difficult.
  - Taking a project from initiation of idea to direct operational stage is a complex and time consuming effort.
  - If these ideas make it to the operational stage, majority of them fail within first six months.
  - Projects involve business operations that differ from Individual business.
  - These operations involve unfamiliar risks.
  - Feasibility study allows groups developing a business idea to preview potential project outcomes and decide if they want to continue developing the project.

## 7.2 Why Do Feasibility Studies? (2/2)

- Though the cost of conducting a feasibility study can seem high, almost always, these costs are relatively minor when compared to the total project cost.
- Small initial expenditure on a feasibility study by a group can help to protect larger capital investments later.
- Feasibility study is a useful tool and is valid for many kinds of projects.

## 7.3 What a Feasibility Study is not? (1 / 2)

- Feasibility studies are conducted on "real-world" projects. Once completed, a study should permit a group to make better decisions for the strategic issues of their specific project.
- A feasibility study is not a business plan. The feasibility study results forms the basis for developing a business plan.
- The purpose of a feasibility study is not to identify new ideas or concepts for a project. These ideas should be clearly identified before a study is initiated.
- Feasibility study should be an objective evaluation of the project's chance for success. Negative results can be just as useful for decision-makers as positive results.

## 7.3 What a Feasibility Study is not? (1 / 2)

- Financers may require a feasibility study before providing loans, but this should not be the only purpose of a study.
- A feasibility study will not determine whether or not a project should be undertaken. The potential members have to decide if the economic returns justify the risks involved in their continuing the project. The results of the feasibility study assist them in this.



## 7.4 Scope of Feasibility Analysis (1 / 4)

Scope of feasibility analysis includes:

- 1. *Need Analysis***- This indicates recognition of a need for the project. The need may affect the organization itself, another organization, the public, or the government. A proposal of how the need may be satisfied is then made. Relevant questions that should be asked include:
  - Is the need significant enough to justify the proposed project?
  - Will the need still exist by the time the project is completed?
  - What are the alternate means of satisfying the need?
  - What are the economic, social, environmental, and political impacts of the need?

## 7.4 Scope of Feasibility Analysis (2/4)

2. *Process Work-* This is the preliminary analysis done to determine what will be required to satisfy the need. The work may be performed by a consultant who is an expert in the project field. The preliminary study often involves system models or prototypes. A simulation of the proposed model can be carried out to predict the outcome before the actual project starts.
3. *Engineering and Design-* This involves a detailed technical study of the proposed project. Written quotations are obtained from suppliers and subcontractors as needed. Technology capabilities are evaluated as needed. Product design, if needed, should be done at this time.

## 7.4 Scope of Feasibility Analysis (3/4)

4. *Cost Estimate*- This involves estimating project cost to an acceptable level of accuracy. Levels of around -5% to +15% are common at this level of a project plan. Both the initial and operating costs are included in the cost estimation. Recurring and nonrecurring costs should also be contained in the cost estimate document.
5. *Financial Analysis*- This involves an analysis of the cash flow profile of the project. The analysis should consider rates of return, inflation, sources of capital, payback periods, breakeven point, residual values, and sensitivity. This is a critical analysis since it determines whether or not and when funds will be available to the project.

## 7.4 Scope of Feasibility Analysis (4/4)

6. *Project Impacts*- This portion of the feasibility study provides an assessment of the impact of the proposed project. Environmental, social, cultural, political, and economic impacts may be some of the factors that will determine how a project is perceived by the public.
7. *Conclusions and Recommendations*- The feasibility study should end with the overall outcome of the project analysis. This may indicate an approval or disapproval of the project.

## 7.5 Elements of a Feasibility Assessment (1/4)

A feasibility report should have the following structure:

- 1. *Executive Summary***- It provides a quick overview of the main points of the assessment, helping to form a picture of the proposal along with the recommendations. It should be concise and include the major findings covered in the main body of the report.
- 2. *Need Analysis***- It analyses the justification of the idea, with a study of possible alternatives. It links the business idea to the current circumstances and helps to inform evaluation of the business idea.
- 3. *Engineering***- Description of the technical aspects of the business idea, including any changes needed to be made to existing processes or the need to add items to existing range of products and services.

## 7.5 Elements of a Feasibility Assessment (2/4)

4. *Advantages and Disadvantages-* Advantages and disadvantages of the business idea compared to alternatives, such as competing products; or for a new concept, its relevance to current practices, and to potential demand.
5. *Market for the Product Offerings-* State the number of customers, expected frequency and size of average purchase, and any reduction in costs across the business arising from the new product or service. Any assumptions about customer purchase behaviour should be identified so that they can be evaluated in terms of likelihood of being achieved or exceeded. Research should focus on: *Customers, Products and Services, Competition, Costing, Suppliers, Resources, Staff etc.*

## 7.5 Elements of a Feasibility Assessment (3/4)

6. *Financial analysis*- find out the profits from a given level of operations, the capital required and how the capital will be found to commence operating.
7. *Risk analysis of the Preferred Solution*- Risk analysis may often include the level of business operation that will ensure that the business does not incur a loss. Sophisticated analysis may consider various business scenarios based on the assumptions made in costing and market analyses.
8. *Comparative Analysis*- Comparative analysis of alternatives should reflect the objectives of the project. The choice between a “high payoff but high risk of failure” option instead of a “low payoff with associated low risk” option is one that you can then make in the context of your objectives, your market and your financial situation.

## 7.5 Elements of a Feasibility Assessment (4/4)

9. *Recommendations-* Recommendations of the preferred alternative with an associated plan of action; or a decision not to proceed, should be covered in this section.



## 7.6 Characteristics of a Feasibility Study

A typical feasibility study checklist might include:

- Summary level
- Evaluate alternatives
- Evaluate market potential
- Evaluate cost effectiveness
- Evaluate producibility
- Evaluate technical base
- A more specific determination of the problem
- Analysis of the state-of-the-art technology
- Assessment of in-house technical capabilities
- Test validity of alternatives
- Quantify weaknesses and unknowns
- Conduct trade-off analysis on time, cost, and performance
- Prepare initial project goals and objectives
- Prepare preliminary cost estimates and development plan

## References:

† Harold Kerzner, *“Project Management, A System Approach to Planning, Scheduling & Controlling”*, 12th Edition, Wiley.

† James. P. Lewis *“Project Planning, Scheduling, and Control”*, 3<sup>rd</sup> Edition, McGraw-Hill Companies.

† Timothy J. Kloppenborg, Vittal Anantatmula, Kathryn N. Wells, *“Contemporary Project Management”* 4<sup>th</sup> Edition, Cengage Learning.

